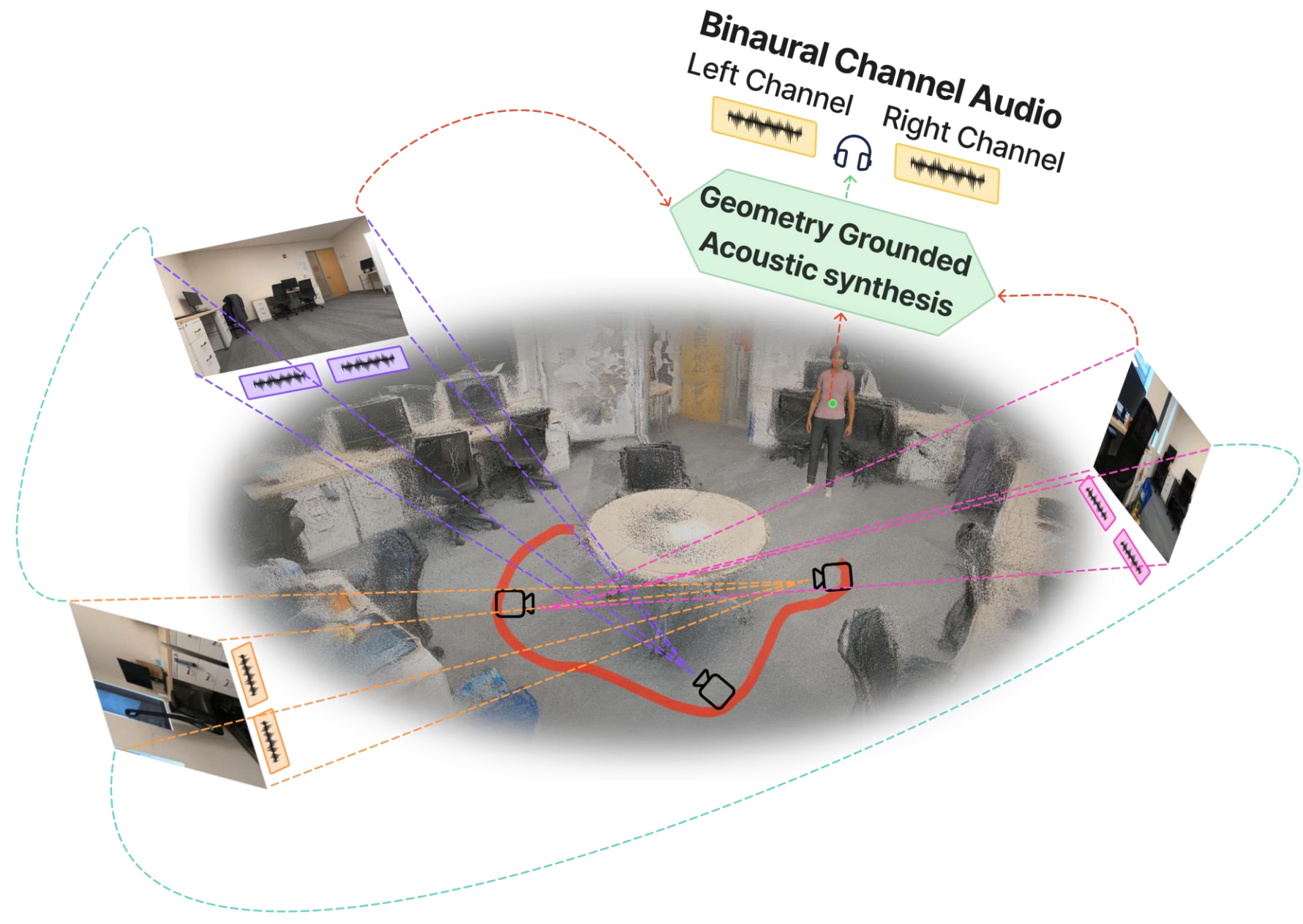


Visual Geometry Grounded Novel-View Acoustic Synthesis

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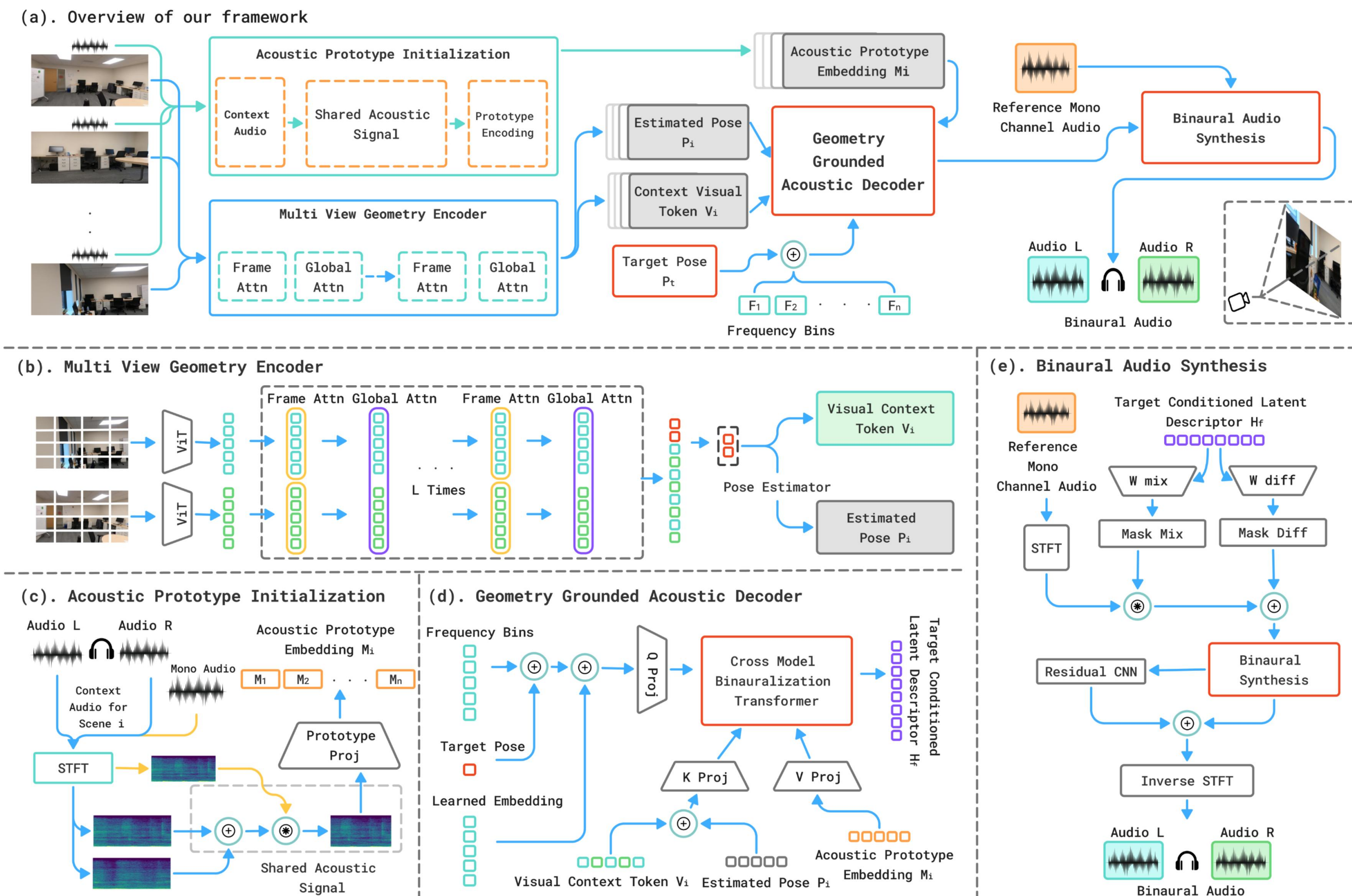
Geometry Grounded Acoustic Synthesis Introduction



- **Novel-view acoustic synthesis (NVAS)** generates spatial audio for unseen listener viewpoints, enabling immersive AR/VR and 3D scene experiences.
- **Existing methods rely on costly 3D reconstruction, target-view rendering, or dense point maps**, while our approach uses feed-forward visual geometry to synthesize binaural audio directly from reference video and audio.

NVAS Pipeline

- **Multi-view geometry encoder** extracts visual context and estimated pose features from sparse reference video frames.
- **Acoustic prototype initialization** encodes reference mono and binaural audio into scene-specific acoustic transfer features.
- **Geometry-Grounded Acoustic Decoder (GGAD)** attends to visual, geometric, and acoustic features to predict target-view binaural transfer fields.
- **Spectral binaural synthesis** applies the predicted transfer fields to the mono audio and reconstructs left-right binaural audio for the queried viewpoint.

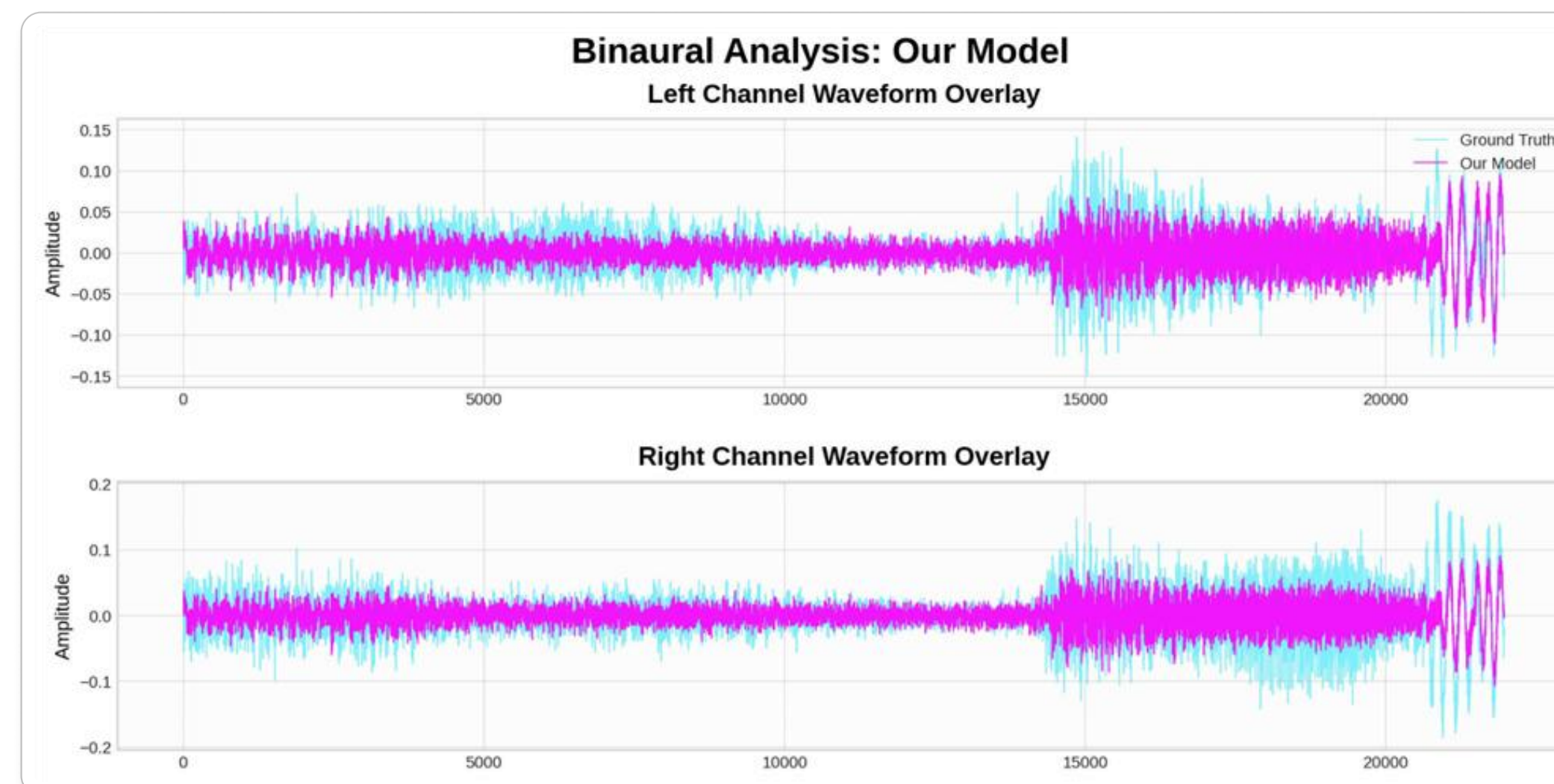
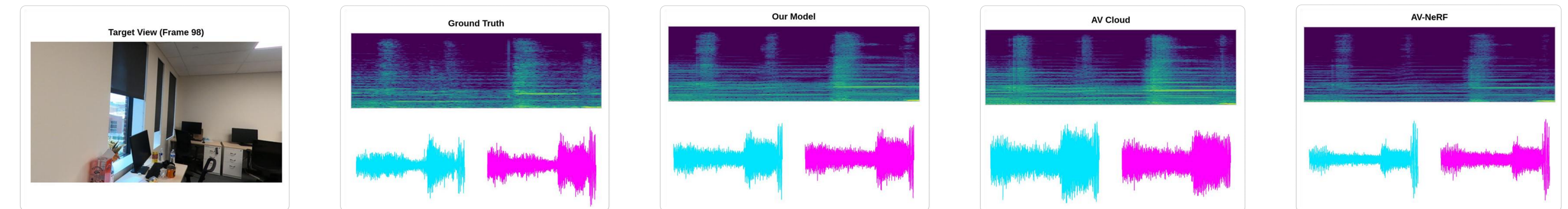


Evaluation and results

RWAVS								
Methods	# Params	FPS	Image	Point	MAG ↓	ENV ↓	LRE ↓	DPAM ↓
Mono-Mono	—	—	✗	✗	1.460	0.445	1.328	0.756
Mono-Energy	—	—	✗	✗	0.532	0.156	1.328	0.510
Stereo-Energy	—	—	✗	✗	0.560	0.160	—	0.535
DSP [8]	163M	—	✗	✗	1.016	0.274	3.468	0.588
VAM [5]	46.7M	174	✓	✗	0.390	0.156	0.996	0.459
ViGAS [6]	13.1M	90	✓	✗	0.370	0.147	1.089	0.357
AVNeRF [15]	12.0M	314	✓	✗	0.370	0.145	1.013	0.381
NACF [16]	0.44M	108	✗	✓	0.459	0.176	1.364	0.506
INRAS [27]	0.31M	475	✗	✓	0.455	0.179	1.503	0.485
NAF [18]	0.22M	261	✗	✓	0.448	0.522	1.204	0.353
AV-Cloud [7]	3.91M	219	✗	✓	<u>0.3652</u>	<u>0.1509</u>	<u>1.0297</u>	<u>0.2776</u>
Our Method	3.24M	189	✗	✗	0.3485	0.1424	0.9589	0.2705

- **Results comparison on RWAVS and Replay-NVAS:** Average metrics are reported using MAG, ENV, LRE, and DPAM, where **lower** values indicate **better** binaural audio synthesis.
- **Our method improves RWAVS performance and shows stronger viewpoint generalization**, outperforming AV-Cloud under both the original random split and the stricter 50/50 train/test split.

Visualizations



- **Qualitative comparison of target-view binaural synthesis**, Spectrogram and waveform results show that our model produces audio closer to the ground truth than AV-NeRF and AV-Cloud.
- **Our method better preserves stereo structure and viewpoint-conditioned audio patterns**, demonstrating more accurate binaural rendering for the queried listener viewpoint.